Radial Basis Function (RBF) Kernel

Example-3

$\begin{pmatrix} u_1 & v_1 \\ u_2 & v_2 \end{pmatrix}$

Normal Model

Calculate $a \cdot b$

$\begin{pmatrix} u_1 & u_2 \\ v_1 & v_2 \end{pmatrix}$

Map Input data from 2-D to infinite Dimension

$\begin{pmatrix} u_1 & v_1 \\ u_2 & v_2 \end{pmatrix}$

Vectors Mapping

$a, b \rightarrow u, v$

Higher Computational Complexity !!!

$u_1 \quad v_1$
$u_2 \quad v_2$
$u_3 \quad v_3$
$u_\ldots \quad v_\ldots$
$u_\ldots \quad v_\ldots$
$u_\infty \quad v_\infty$

Kernelized Model

Calculate $u \cdot v$

$\begin{pmatrix} u_1 & u_2 & u_3 & \ldots & \ldots & u_\infty \\ v_1 & v_2 & v_3 & v_\ldots & v_\ldots & v_\infty \end{pmatrix}$

Higher Computational Complexity !!!